

INDUSTRIAL AI MONITORING SYSTEM

TECHNICAL DOCUMENTATION REPORT

Tennessee Eastman Process Telemetry

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DEPLOYMENT TARGET: Enterprise Control Room Panel

SYSTEM VERSION: 3.1.0-Premium

1. Project Overview & Core Abstract

The Industrial AI Monitoring System represents a state-of-the-art predictive telemetry console engineered for safety-critical plant environments. Using the classical Tennessee Eastman Process (TEP) benchmark chemical plant simulator, the project builds a multi-tiered diagnostic intelligence suite.

Traditionally, refinery operators face highly complex, high-dimensional alarm states that are difficult to diagnose rapidly. Our system addresses this by integrating real-time Classification (predicting active fault indexes), ~~Anomaly Detection (distinguishing standard noise from structural drifts)~~ and Deep Learning LSTM Sequence Trajectory Predictors to forecast process failures early. All mathematical insights are rendered on a high-fidelity dark cybernetic telemetry panel running at a simulated 5 frames per second control loop.

2. Model Performance Summary

To maximize diagnostic accuracy, three distinct ML/DL model types were evaluated. Below is the comparative validation summary:

Model Type / Architecture	Classification Accuracy
Static Random Forest (Static RF)	72.00%
Temporal Random Forest (Temporal RF)	96.10%
Deep Sequential LSTM (Temporal LSTM)	97.66%

*Note: Static RF struggles to capture transitional states because it evaluates measurements independent of history. Temporal RF and LSTM models solve this by utilizing multi-frame sliding window buffers, bringing diagnostic rates to near-perfect levels.

3. Project Workspace File Structure

/ (Root Folder)

- download_dataset.py: Script that automatically downloads the base TEP subset (tep_subset.csv).
- download_models.py: Script that automatically retrieves pre-trained network model binaries.
- prepare_replay_data.py: Sorts the datasets temporally and prepares the 10,000-frame TEP replay data.
- train_tep_model.py: Trains the multiclass Random Forest fault classifier.
- train_anomaly_detector.py: Fits the Isolation Forest anomaly detector on baseline data.
- train_lstm_model.py: Configures and trains the deep sequence LSTM neural network weights.
- create_temporal_features.py: Extracts rolling mean, std, and delta parameters over window buffers.
- train_temporal_model.py: Trains a Random Forest on extracted temporal features.
- requirements.txt: Defines external dependencies (TensorFlow, scikit-learn, joblib, plotly, streamlit).
- README.md: Comprehensive markdown configuration guide.

/app/ (User Interface)

- dashboard.py: The premium HMI control room dashboard built on Streamlit with deep Custom CSS styling.

/data/ (Telemetry Storage)

- tep_subset.csv: Base Tennessee Eastman dataset.
- tep_replay.csv: Ordered fault replay file prepared by prepare_replay_data.py.
- presets.json: Telemetry presets mapping to standard/abnormal states.
- incident_logs.csv: Autonomous logging registry for recorded critical faults.
- feature_importances.csv & pca_background.csv: Support data for visual rendering panels.

4. Advanced Control Panel Systems

A. Root Cause Analysis (RCA) Engine

To resolve the operational challenge of rapid fault localization, the system computes the exact deviation values between the live active sensor measurements and a cached normal operating baseline. It dynamically ranks these deviations and plots them to instantly highlight abnormal process variables.

B. Spatial Telemetry Heatmap

The 52 process variables are projected into a physical 4 x 13 spatial control grid representation. Using Plotly Express turbo-scaling, the layout represents the actual plant flow. Crimson glows instantly pinpoint highly perturbed process sectors during pre-failure states.

C. Predictive Warning Early Alert

Monitors rising confidence trajectories inside a sliding window. If a continuous positive trend is detected and the prediction exceeds a 60% probability threshold, the panel triggers an early predictive maintenance alert, allowing operators to halt plant processes before catastrophic failure.

D. High-Speed HMI Replay Loop

Features sidebar selection sliders and sequential replay engines. The UI autorefresh loop runs at 5 frames per second server-side, bringing a high-speed telemetry experience identical to real-world HMIs.